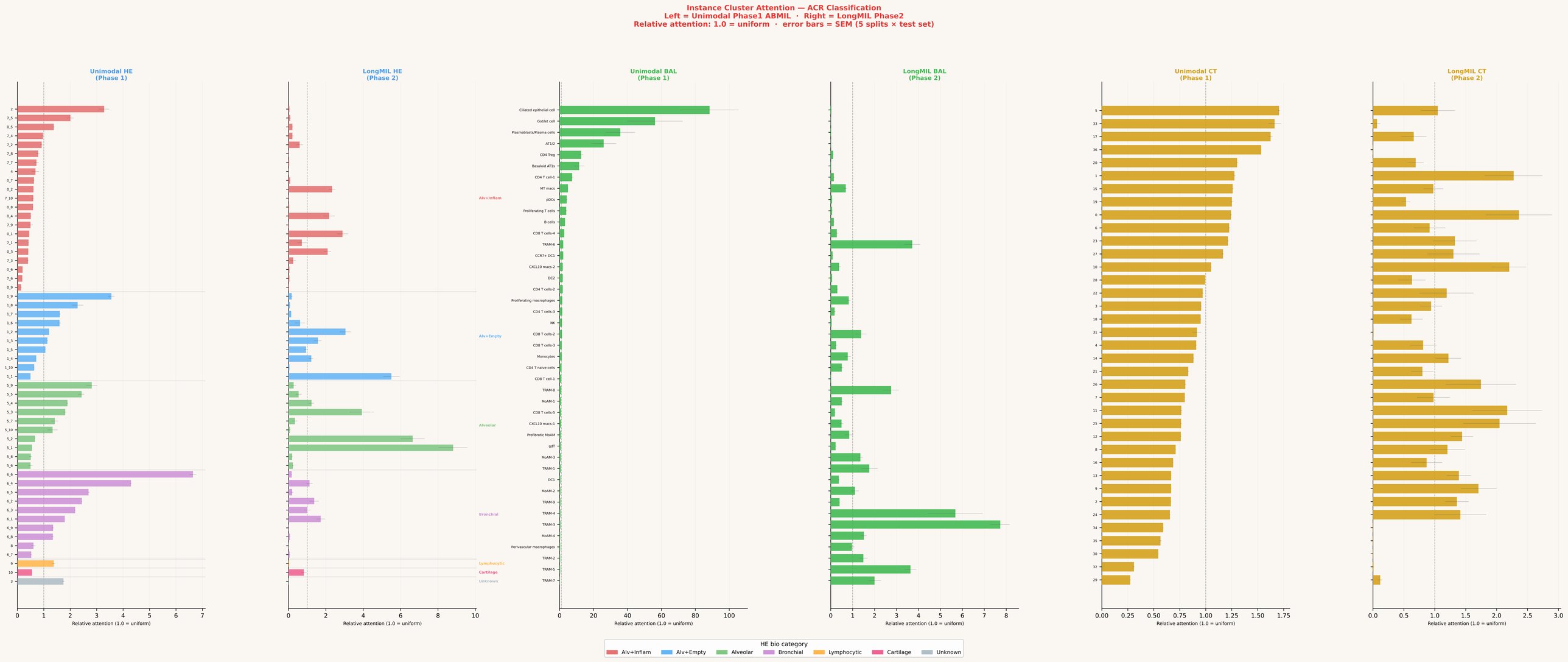
Instance Cluster Attention — ACR Classification
Left = Unimodal Phase1 ABMIL · Right = LongMIL Phase2
Relative attention: 1.0 = uniform · error bars = SEM (5 splits × test set)
Unimodal HE (Phase 1)
LongMIL HE (Phase 2)
Unimodal BAL (Phase 1)
LongMIL BAL (Phase 2)
Unimodal CT (Phase 1)
LongMIL CT (Phase 2)
Alv+Inflam
Alv+Empty
Alveolar
Bronchial
Lymphocytic
Cartilage
Unknown
Ciliated epithelial cell
Goblet cell
Plasmablasts/Plasma cells
AT1/2
CD4 Treg
Basaloid AT1s
CD4 T cell-1
MT macs
pDCs
Proliferating T cells
B cells
CD8 T cells-4
TRAM-6
CCR7+ DC1
CXCL10 macs-2
DC2
CD4 T cells-2
Proliferating macrophages
CD4 T cells-3
NK
CD8 T cells-2
CD8 T cells-3
Monocytes
CD4 T naive cells
CD8 T cell-1
TRAM-8
MoAM-1
CD8 T cells-5
CXCL10 macs-1
Profibrotic MoAM
gdT
MoAM-3
TRAM-1
DC1
MoAM-2
TRAM-9
TRAM-4
TRAM-3
MoAM-4
Perivascular macrophages
TRAM-2
TRAM-5
TRAM-7
Relative attention (1.0 = uniform)
HE bio category
Alv+Inflam
Alv+Empty
Alveolar
Bronchial
Lymphocytic
Cartilage
Unknown